THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL MCQ TEST SET 1

Full Syllabus (Chapters 1–14) | RBSE Pattern | Session 2026–27

Time: 1 Hour Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–50 are Multiple Choice Questions (1 mark each); there is no negative marking.
3. Choose the correct option and write its letter (a, b, c or d) against the question number.
4. Questions cover the whole syllabus of the class.
5. Medium level: multi-step questions.

MULTIPLE CHOICE QUESTIONS — 50 × 1 = 50

Choose the correct option and write it in your answer book.

1. The smallest number by which 72 must be multiplied to get a perfect square is: 1

(a) 4 **(b)** 1 **(c)** 2 **(d)** 3

2. $\dfrac{3^6 \times 3^4}{3^5}$ equals: 1

(a) 3^{19} **(b)** 3^5 **(c)** 3^{15} **(d)** 3^7

3. The decimal value of 10110_2 is: 1

(a) 22 **(b)** 24 **(c)** 20 **(d)** 28

4. The angles of a quadrilateral are x°, $(x+10)^\circ$, $(x+20)^\circ$ and $(x+30)^\circ$. The value of x is: 1

(a) 65 **(b)** 85 **(c)** 76 **(d)** 75

5. The digit that replaces * in 7*93 so that the number is divisible by 11 is: 1

(a) 3 **(b)** 5 **(c)** 8 **(d)** 2

6. $(3x+2)(2x+5)$ equals: 1

(a) $6x^2 - 19x + 10$ **(b)** $6x^2 + 19x - 10$ **(c)** $6x^2 + 19x + 10$ **(d)** $6x^2 + 10$

7. Two numbers are in the ratio 4 : 7 and their sum is 99. The larger number is: 1

(a) 63 **(b)** 36 **(c)** 49 **(d)** 72

8. An article is sold for ₹ 1260 at a profit of 5%. Its cost price (in ₹) is: 1

(a) 1200 **(b)** 1197 **(c)** 1080 **(d)** 1320

9. An isosceles triangle has equal sides 30 cm and base 36 cm. Its height (in cm) is: 1

(a) 25 **(b)** 12 **(c)** 24 **(d)** 23

10. If A:B = 3:4 and B:C = 8:9, then A:C is: 1

(a) 2:3 **(b)** 27:32 **(c)** 3:5 **(d)** 3:9

11. A prism has 18 edges. The number of its faces is: 1

(a) 7 **(b)** 9 **(c)** 6 **(d)** 8

12. The mean mark of 10 students is 40 and of another 15 students is 60. The mean of all 25 students is: 1

(a) 54 **(b)** 1300 **(c)** 52 **(d)** 50

13. On expanding $3(x+4)+2x$ we get: 1

(a) $5x+4$ **(b)** $3x+14$ **(c)** $5x+12$ **(d)** $6x+12$

14. An L-shaped figure is made by removing a 8 cm by 6 cm rectangle from a 20 cm by 15 cm rectangle. Its area is: 1

(a) 300 cm² **(b)** 108 cm² **(c)** 348 cm² **(d)** 252 cm²

15. The smallest number by which 192 must be divided to get a perfect cube is: 1

(a) 4 **(b)** 3 **(c)** 5 **(d)** 6

16. The value of $\left(\frac{2}{3}\right)^{-2}$ is: 1

(a) $\frac{9}{4}$ **(b)** $-\frac{9}{4}$ **(c)** $\frac{4}{9}$ **(d)** 3

17. The binary form of 45 is: 1

(a) 110001 **(b)** 101101 **(c)** 101100 **(d)** 101110

18. The adjacent sides of a parallelogram are in the ratio 3:5 and its perimeter is 64 cm. The longer side is: 1

(a) 32 cm **(b)** 20 cm **(c)** 12 cm **(d)** 40 cm

19. The smallest digit that replaces * in 52*8 so that the number is divisible by 6 is: 1

(a) 1 **(b)** 2 **(c)** 5 **(d)** 0

20. $(2x - 3y)^2$ equals: 1

(a) $4x^2 - 12xy + 9y^2$ **(b)** $4x^2 - 6xy + 9y^2$ **(c)** $4x^2 - 9y^2$ **(d)** $4x^2 + 12xy + 9y^2$

21. Two numbers are in the ratio 3 : 5. If 4 is added to each, the numbers become 28 and 44. The original smaller number is: 1

(a) 24 **(b)** 48 **(c)** 20 **(d)** 28

22. A watch is sold for ₹ 1530 at a loss of 15%. Its cost price (in ₹) is: 1

(a) 1980 **(b)** 1620 **(c)** 1759.5 **(d)** 1800

23. A rectangle has a diagonal of 30 cm and one side 18 cm. Its perimeter (in cm) is: 1

(a) 42 **(b)** 84 **(c)** 96 **(d)** 86

24. 12 workers working 8 hours a day finish a job in 15 days. 10 workers working 6 hours a day will finish it in (days): 1

(a) 18 **(b)** 24 **(c)** 180 **(d)** 26

25. A pyramid has 7 vertices. The number of its edges is: 1

(a) 14 **(b)** 12 **(c)** 8 **(d)** 6

26. The mean of 59, 36, 46, 44, 50, 11 is 41. If 46 is replaced by 52, the new mean is: 1

(a) 41 **(b)** 49.67 **(c)** 42 **(d)** 47

27. $(x + 2)(x + 5)$ equals: 1

(a) $x^2 + 7x + 10$ **(b)** $x^2 + 7$ **(c)** $x^2 + 10x + 7$ **(d)** $2x + 7$

28. A parallelogram has base 18 cm and height 10 cm. If the base is changed to 15 cm (same area, other side), the corresponding height (in cm) is: 1

(a) 12 **(b)** 13 **(c)** 165 **(d)** 15

29. The ones digit of 127^2 is: 1

(a) 4 **(b)** 1 **(c)** 7 **(d)** 9

30. If $3^x \times 3^4 = 3^{11}$, then x is: 1

(a) 15 **(b)** 7 **(c)** 44 **(d)** 9

31. XLIV + LXVII equals: 1

(a) CXI **(b)** CXII **(c)** CI **(d)** CXXI

32. The diagonals of a parallelogram ABCD meet at O. If AO $= (2x + 3)$ cm and OC $= (3x - 4)$ cm, then AC is: 1

(a) 7 cm **(b)** 34 cm **(c)** 17 cm **(d)** 36 cm

33. One step of the Kaprekar routine applied to 1895 gives: 1

(a) 8163 **(b)** 8280 **(c)** 8262 **(d)** 8361

34. The value of $63^2 - 37^2$ is: 1

(a) 2600 **(b)** 676 **(c)** 10000 **(d)** 5338

35. A mixture of 60 L has milk and water in the ratio 7 : 3. The quantity of water is: 1

(a) 42 L **(b)** 18 L **(c)** 13 L **(d)** 23 L

36. After a discount of 15%, a pair of shoes is sold for ₹ 1530. The marked price (in ₹) is: 1

(a) 1620 **(b)** 1759.5 **(c)** 1800 **(d)** 1980

37. The side of a rhombus is 30 cm and one diagonal is 36 cm. The other diagonal (in cm) is: 1

(a) 24 **(b)** 48 **(c)** 49 **(d)** 50

38. The ratio of boys to girls in a school is 7:8. If there are 1350 students, the number of girls is: 1

(a) 720 **(b)** 1335 **(c)** 630 **(d)** 675

39. All 8 faces of a polyhedron are triangles. The number of its edges is: 1

(a) 16 **(b)** 8 **(c)** 24 **(d)** 12

40. The median of 46, 96, 29, 15, 31, 39, 70, 52 is: 1

(a) 40 **(b)** 47.25 **(c)** 42.5 **(d)** 31

41. If $a = 9$ and $b = 4$, the value of $3a - 5b + 8$ is: 1

(a) 30 **(b)** 14 **(c)** 16 **(d)** 15

42. A rhombus has side 15 cm and one diagonal 18 cm. Its area is: 1

(a) 108 cm² **(b)** 270 cm² **(c)** 60 cm² **(d)** 216 cm²

43. $\sqrt{1.44}$ is equal to: 1

(a) 1.3 **(b)** 0.12 **(c)** 1.2 **(d)** 12.0

44. If $(2^3)^x = 2^{12}$, then x is: 1

(a) 36 **(b)** 4 **(c)** 15 **(d)** 9

45. The largest 3-digit number in base 2, written in decimal, is: 1

(a) 5 **(b)** 3 **(c)** 7 **(d)** 8

46. In a rhombus ABCD, $\angle A = (4x + 10)^\circ$ and $\angle B = (5x + 17)^\circ$. The value of x is: 1

(a) 21 **(b)** 18 **(c)** 16 **(d)** 17

47. The smallest number divisible by 4, 6 and 10 is: 1

(a) 30 **(b)** 64 **(c)** 60 **(d)** 120

48. The value of $(x + 3)^2$ when $x = 7$ is: 1

(a) 79 **(b)** 106 **(c)** 100 **(d)** 58

49. 6 pipes fill a tank in 10 hours. 4 such pipes will fill it in: 1

(a) 60 hours **(b)** 12 hours **(c)** 18 hours **(d)** 15 hours

50. The simple interest on a sum at 5% p.a. for 4 years is ₹ 600. The sum (in ₹) is: 1

(a) 3300 **(b)** 3000 **(c)** 6000 **(d)** 2700

END OF QUESTION PAPER **Total: 50 Marks**

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